

FORT GRATIOT LH, MI, USA

WMO: 997258

Lat: 43.0063N

Lon: 82.4224W

Elev: 591

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.8	(c) 5.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 32.5	(k) 23.3	(l) 27.2	(m) 27.0	(n) 10.6	(o) 220	(p) 0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.2	(c) 88.0	(d) N/A	(e) 84.9	(f) N/A	(g) 81.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 10.5	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 30.4	(b) 25.1	(c) 20.9	(e) DB	(f) -4.0	(g) 93.6	(h) 7.0	(i) 3.8	(j) -9.0	(k) 96.3	(l) -13.1	(m) 98.6	(n) -17.0	(o) 100.7	(p) -22.0	(q) 103.5	(r) N/A
(4) 30.4	(5) 25.1	(6) 20.9	(7) DB	(8) -4.0	(9) 93.6	(10) 7.0	(11) 3.8	(12) -9.0	(13) 96.3	(14) -13.1	(15) 98.6	(16) -17.0	(17) 100.7	(18) -22.0	(19) 103.5	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	23.6	23.9	33.3	44.6	56.0	65.2	70.7	69.8	63.6	52.2	40.0	31.6		
	DBStd	18.65	10.65	9.91	9.99	8.74	8.77	7.56	6.22	5.36	6.90	7.84	8.64	8.39		
	HDD50	3277	819	730	528	211	35	1	0	0	1	63	317	572		
	HDD65	6755	1283	1150	983	611	307	91	18	16	106	403	751	1036		
	CDD50	2551	1	0	10	50	220	455	642	613	409	133	17	1		
	CDD65	554	0	0	0	0	27	95	195	164	65	8	0	0		
	CDH74	3593	0	0	2	25	239	710	1325	898	341	53	0	0		
	CDH80	1023	0	0	0	3	57	222	420	219	86	16	0	0		
	WSAvg	10.2	10.8	10.9	11.7	12.0	10.2	9.5	8.7	8.8	9.1	10.5	10.5	10.4		
	PrecAvg	34.1	1.9	2.0	2.6	3.0	2.7	3.1	2.7	3.0	3.6	2.5	3.4	2.9		
	PrecMax	43.4	3.3	4.3	5.7	5.2	4.8	5.3	4.3	7.1	9.9	4.7	5.3	4.6		
(15) Precipitation	PrecMin	25.2	0.9	0.6	1.2	1.5	0.9	0.8	1.2	0.8	0.5	0.7	0.7	1.3		
	PrecStd	4.5	0.8	1.1	1.1	0.9	1.1	1.1	1.0	2.0	2.4	1.1	1.4	0.8		
	DB	54.4	53.3	71.5	77.8	86.1	91.0	93.7	89.0	88.3	80.1	66.3	57.1			
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	48.1	47.3	62.3	71.8	81.3	86.5	88.6	85.4	81.9	72.3	59.9	51.0			
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% DB	42.9	41.1	55.1	65.7	76.5	81.8	84.6	82.4	77.6	68.1	56.4	46.9			
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10% DB	38.1	36.8	48.0	59.8	71.5	77.6	81.3	79.1	74.3	64.2	53.0	42.9			
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MDBR	12.0	12.2	13.4	14.8	15.8	14.9	14.2	13.7	14.4	13.6	11.7	9.9			
	5% DB	14.9	16.7	23.3	26.5	23.7	21.3	19.9	18.0	18.3	17.7	17.3	14.5			
(35) Mean Daily Temperature Range	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub	0.309	0.320	0.338	0.383	0.408	0.432	0.420	0.409	0.377	0.358	0.319	0.294			
	taud	2.338	2.302	2.371	2.275	2.261	2.222	2.282	2.300	2.376	2.405	2.485	2.445			
	Ebn at Noon	259	277	287	280	275	267	270	269	270	260	257	257			
	Edh at Noon	26	32	34	41	42	44	41	39	34	29	23	22			
(44) All-Sky Solar Radiation	RadAvg	416	681	1094	1445	1701	1930	1902	1654	1324	843	504	344			
	RadStd	54	69	110	126	145	108	101	76	116	75	65	33			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page